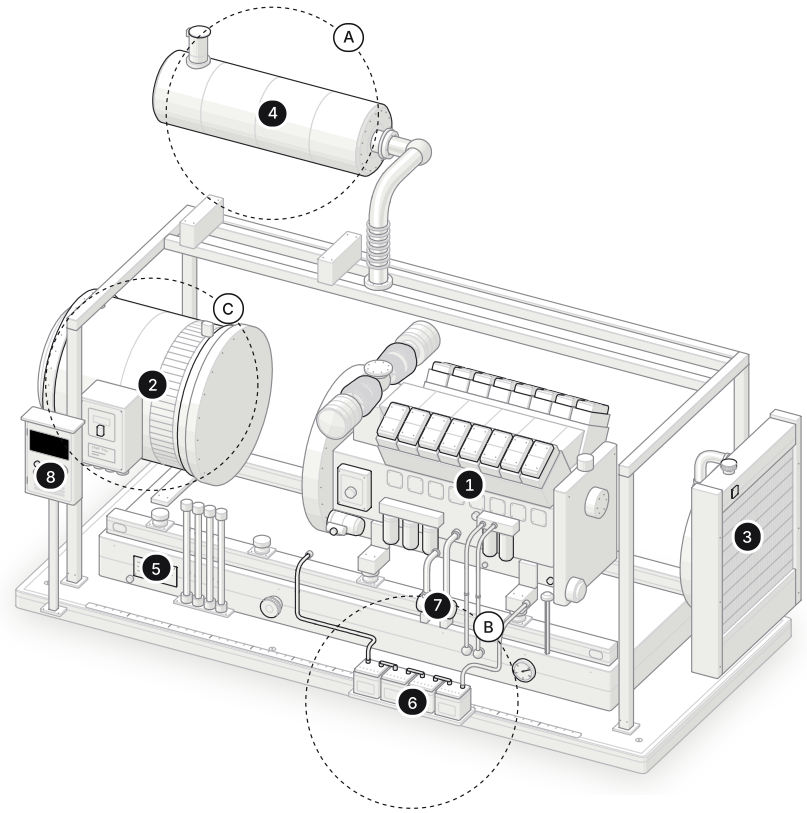
Standby generator



What happens when a data centre loses its mains power?

The UPS batteries carry the servers for the first seconds. Within 8.8 s the diesel generator has cranked, fired, reached 1,500 rpm and 50 Hz, and the transfer switch has moved the load across to it.

2,500 kVA
2,000 KW AT 0.8 POWER FACTOR

1,500 rpm
16-CYLINDER V DIESEL, 50 HZ

8.8 s
MAINS FAILURE TO ON LOAD

5,500 L
BASE TANK, 13 H AT 1,600 KW

PARTS

01 **Diesel engine**
V16, twin turbos; turns the alternator at 1,500 rpm.

02 **Alternator**
Makes 400 V, 50 Hz power; breaker and cables in its terminal box.

03 **Radiator and fan**
Sheds the engine's heat to the air of the generator hall.

04 **Exhaust silencer**
Quietens the exhaust; the rain cap lifts when gas flows.

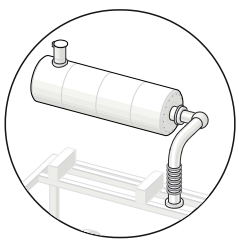
05 **Base fuel tank and mounts**
5,500 L of diesel under the skid; rubber mounts stop vibration.

06 **Starter batteries**
Four 12 V blocks give 24 V to crank the engine.

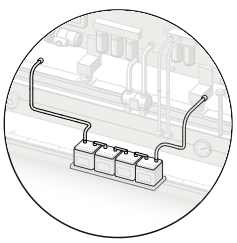
07 **Jacket-water heater**
Keeps the coolant at 40 °C so the set starts fast.

08 **Control panel**
Senses the mains failure, starts, watches and protects the set.

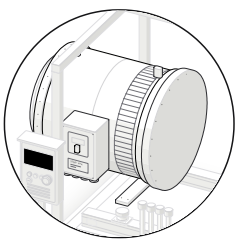
CLOSE-UPS



(A) **Turbos and exhaust**
Twin turbochargers feed one pipe through a bellows to the roof silencer.

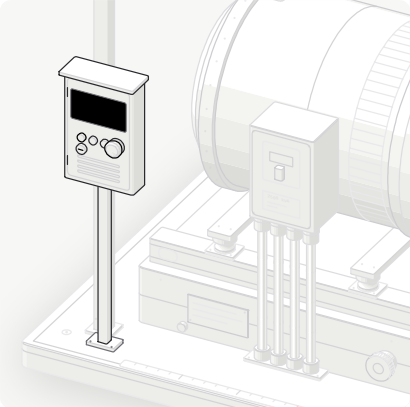


(B) **Starter batteries**
Four 12 V blocks in a rack give 24 V to the starter motor.



(C) **Alternator**
Bolted to the flywheel housing; its cables drop into a trench to the switchgear.

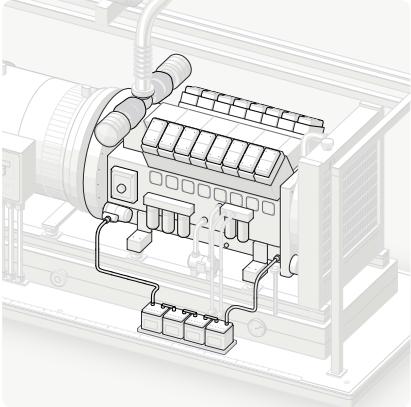
How it works



01

Mains fails

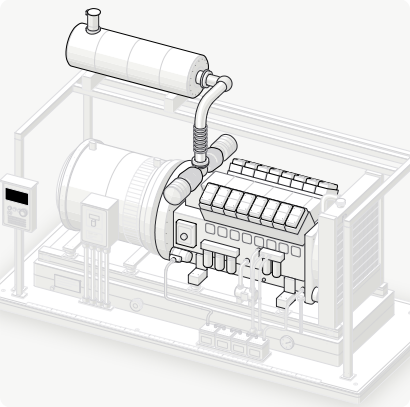
The UPS carries the hall. The controller waits 1 s to be sure the mains is really gone.



02

Batteries crank

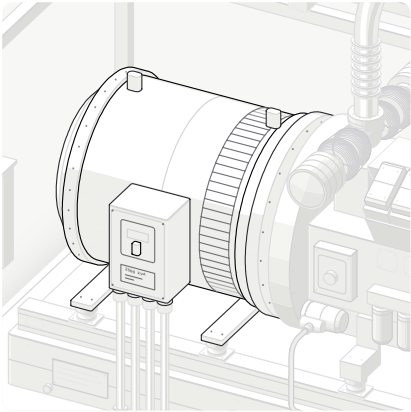
The 24 V batteries drive the starter motor, turning the engine at about 150 rpm.



03

Engine fires

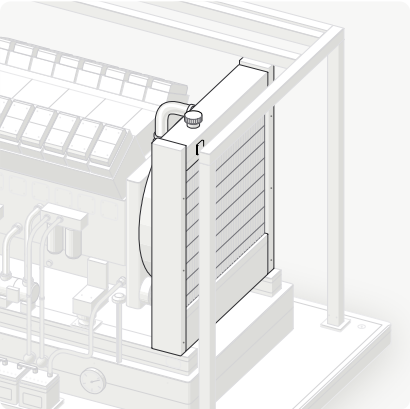
Diesel ignites under compression and the engine runs up to 1,500 rpm. With four poles that is 50 Hz.



04

On load

At 8.8 s the transfer switch moves 1,600 kW onto the alternator. The speed dips, then the governor catches it.



05

Cooling the engine

The radiator and fan carry the engine's heat away for as long as it runs.

WHY KEEP IT WARM?

A cold diesel starts slowly and can't take a big load at once. The jacket-water heater holds the engine at 40 °C around the clock, so it can fire within seconds and take 1,600 kW in one step.

SPECIFICATION

Rating	2,500 kVA / 2,000 kW	Transfer	8.8 s (limit 10 s)
Output	400 V, 50 Hz, 3,608 A a phase	Fuel	5,500 L, about 424 L/h at 1,600 kW
Engine	V16 diesel, 1,500 rpm	Standby	coolant held at 40 °C
Start	24 V electric, about 150 rpm		